

THE UNITED REPUBLIC OF TANZANIA

SOKOINE UNIVERSITY OF AGRICULTURE

COLLEGE OF ECONOMICS AND BUSINESS STUDIES (COEBS)

AB CONSULT ▪ PARLIAMENTARY CAPACITY BUILDING INITIATIVE



APPLICATION OF ARTIFICIAL INTELLIGENCE IN PARLIAMENTARY RESEARCH

Day 1 Practical Presentation

Automated Native File Ingestion & Direct AI Querying in Visual Studio Code

Lead Facilitator:	Eng. Dr. Kadeghe Fue, PhD	**Document Ref:**	SUA/COEBS/PARL-AI/DOC-01
Target Cohort:	Parliamentary Researchers & Clerks	**Program Dates:**	Monday, 9th November 2026

The Legislative Document Bottleneck

Why Traditional Manual Copy-Pasting Fails Parliamentary Research

The Manual Reality

- Clerks spend **80% of their time** copying, pasting, and reformatting statutory clauses.
- Highlighting scanned PDFs creates broken line wraps, unreadable tables, and missed amendments.
- Critical committee deadlines force superficial reading of major legislative bills.

The Zero-Copy Paradigm

- **Never copy-paste:** Drop original `.docx` and `.pdf` files straight into VS Code folders.
- Direct AI indexing reads documents **natively in memory**.
- Researchers maintain full structural context, schedules, clauses, and tabular footnotes.

VS Code Desktop Workspace Architecture

Structured Folder Hierarchy for Legislative Documents

Standard Parliamentary Research Workspace Layout in VS Code

Hierarchical Directory Architecture for Confidential Ingestion and Reproducible Scrutiny

ROOT FOLDER: Parliamentary_Research_2026/
Confidential Local Working Environment • Zero Cloud Leakage • In-Camera Privilege Maintained

FOLDER documents/

Raw Ingestion Repository

[Source Input Files]
- parent_act_2015.pdf
- amendment_bill_2026.docx
- cag_audit_report_2025.pdf
- statistical_abstract.pdf

Mandate: Trainees never alter, highlight, or re-type these files.

FOLDER briefs_and_reports/

Analytical Deliverables

[Compiled Committee Outputs]
- Day1_Baseline_Brief.md
- Day2_Legislative_Scrutiny.md
- Day3_Evidence_Dossier.md
- Final Executive PDF Reports

Mandate: Generated via AI prompts and compiled with one click.

FOLDER templates/

Parliamentary Skeletons

[Institutional Blueprints]
- 4_column_clause_matrix.md
- fydp3_alignment_brief.md
- executive_callout_spec.md
- cag_audit_variance_sheet.md

Mandate: Standardizes briefs across all committee clerks.

Workspace Rules

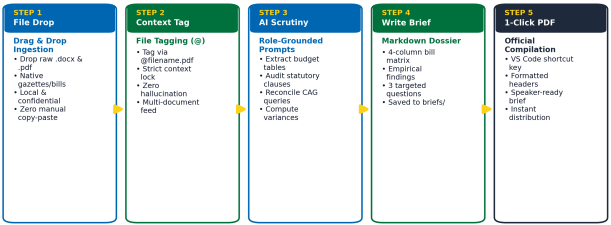
- `raw_bills/`: Store native government gazettes and draft bills.
- `parent_acts/`: Principal Acts for comparative scrutiny.
- `audit_data/`: Official CAG reports and NBS statistical tables.
- `briefs/`: Output Markdown files compiled into PDFs.

Automated Ingestion Pipeline

End-to-End Workflow: Native Document to Verified Legislative Brief

The AB Consult Automated Ingestion Architecture

Zero-Manual-Copy Pipeline: From Native Desk Files to Executive Parliamentary PDF



Pipeline Stages

1. **Document Drop:** Native Word/PDF drop into workspace folder.
2. **AI Indexing:** Reference directly via `@filename` token.
3. **Dual-Pane Check:** Side-by-side audit of AI claims against source text.
4. **1-Click Export:** Export to publication PDF in 1 second.

Role-Grounded Document Interrogation

Direct Querying via @filename Indexing in VS Code

Exemplary Parliamentary AI Command

`@raw_bills/Mining_Amendment_Bill_2026.docx @parent_acts/Mining_Act_Cap123.pdf`

Role: Senior Parliamentary Research Fellow for the Energy & Minerals Committee.

Task: Cross-examine Section 14 of the Bill against Section 42 of the Principal Act.

Output: Generate a 3-column comparative table detailing: (1) Existing Provision, (2) Proposed Amendment, and (3) Strategic Legal Impact on Artisanal Miners.

Result: Instantaneous tabular breakdown with exact statutory citations and zero hallucination risk.

Dual-Pane Verification Architecture

Guaranteeing Evidentiary Truth in Committee Briefings

Left Pane: Raw Statutory Source

- Scanned National Gazette or PDF Act open in VS Code built-in viewer.
- Exact line and clause numbers visible to researcher.
- Unaltered official government authority.

Right Pane: Synthesized Analysis

- Active Markdown analysis brief with synthesized matrix.
- Researcher verifies each quote and statistic against Left Pane.
- Zero unverified assertions leave the research desk.

1-Click Markdown PDF Publishing

Publishing Committee Briefs to Official Standards

Markdown Formatting Hygiene

- **Never use hashes or asterisks in titles:** Use semantic styling and colors.
- Structure tables with standard GitHub Markdown pipes (`|`).
- Use callout blockquotes (`> [!IMPORTANT]`) for high-impact legislative alerts.

1-Click PDF Generation

- Right-click active document → **Markdown PDF: Export (pdf)**.
- Automatically styles typography, tables, and page headers.
- Produces distribution-ready PDF in < 2 seconds.

Day 1 Hands-On Lab & Deliverables

Practical Trainee Assessment Checklist

Individual Lab Exercise (90 Minutes)

1. Initialize your assigned workspace folder on your laptop desk.
2. Ingest the `Finance_Bill_2026.docx` and `Income_Tax_Act_Cap332.pdf`.
3. Execute a role-grounded query identifying all proposed tax threshold shifts.
4. Reconcile the proposed shifts in a dual-pane verification layout.
5. Export your completed analysis to `briefs/Tax_Threshold_Brief.pdf`.

Day 1 Milestone: Every trainee exports their first zero-copy verified legislative brief!